



LAB ASSIGNMENT -3
OBJECT ORIENTED PROGRAMMING (MCA -302)

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Course – MCA (26-28)

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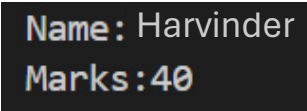
1. Create a class Student with private data members name and marks. Write a friend function to access and display these private data members.

```
#include<iostream>
using namespace std;
class Student{
private:
    string name;
    int marks;
public:
    void setData(string n,int m){
        name=n;
        marks=m;
    }
    friend void display(Student s);
};

void display(Student s){
    cout<<"Name:"<<s.name<<endl;
    cout<<"Marks:"<<s.marks<<endl;
}

int main(){
    Student s;
    s.setData("Harvinder",40);
    display(s);
    return 0;
}
```

OUTPUT:



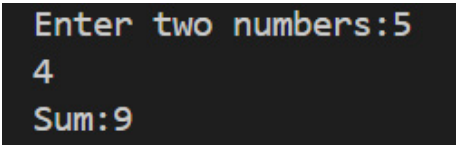
```
Name: Harvinder
Marks: 40
```

2. Create a class Number with two private data members a and b. Use a friend function to access these members and calculate and display their sum.

```
#include<iostream>
using namespace std;
class      Number{
private:
int a,b;
public:
void input(){
    cout<<"Enter two numbers:";
    cin>>a>>b;
}

friend void sum(Number n);
};
void sum(Number n){
cout<<"Sum:"<<n.a+n.b;
}
int main(){
    Number n;
    n.input();
    sum(n);
    return 0;
}
```

OUTPUT:

A screenshot of a terminal window with a dark background. It shows the output of the C++ program: "Enter two numbers:5", "4", and "Sum:9". The numbers 5 and 4 are on separate lines, and "Sum:9" is on a third line.

```
Enter two numbers:5
4
Sum:9
```

3. Create a class Employee with a private data member salary. Create two objects and use a friend function to compare their salaries and display the higher salary.

```
#include<iostream>
```

```

using namespace std;

class Employee{
private:
int salary;
public:
void input(){
    cin>>salary;
}

friend void compare(Employee e1,Employee e2);
};

void compare(Employee e1, Employee e2){
    if(e1.salary>e2.salary){
        cout<<"Highest Salary:"<<e1.salary;
    }
    else if(e2.salary>e1.salary){
        cout<<"Highest Salary:"<<e2.salary;
    }
}

int main(){
    Employee e1,e2;
    cout<<"Enter salary of Employee 1:\n";
    e1.input();
    cout<<"Enter salary of Employee 2:\n";
    e2.input();
    compare(e1,e2);
    return 0;
}

```

OUTPUT:

```
Enter salary of Employee 1:
6000
Enter salary of Employee 2:
8000
Highest Salary:8000
```

4. Create a class Student with a static data member to keep track of the number of objects created. Create three objects and display the total number of objects.

```
#include<iostream>
using namespace std;
class Student{
private:
    static int count;
public:
    Student(){
        count++;
    } static void display(){ cout<<"Total number of
objects created:"<<count;

}

};

int Student::count=0;

int main(){
    Student s1;
    Student s2;
    Student s3;
    Student::display();
    return 0;
}
```

OUTPUT:

```
Total number of objects created:3
```

5. Create a class BankAccount with:

Private data members:

- accountNo
- balance

Static data member:

- totalAccounts

Implement:

- A constructor to initialize the account details.
- A friend function to display the private account details.
- A static member function to display the total number of accounts created.

Create at least three objects and demonstrate both the friend function and static member function.

```
#include<iostream>
using namespace std;
class BankAccount{
private:
    int account_no;
    float balance;
    static int totalAccounts;

public:
    BankAccount(int acc,float bal){
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```

```

        account_no=acc;
        balance=bal;
        totalAccounts++;
    }

friend void displayAccount(BankAccount b);

static void displaytotalAccounts(){
    cout<<"Total accounts created:"<<totalAccounts;
}

};

int BankAccount::totalAccounts=0;

void displayAccount(BankAccount b){
    cout<<"Enter Account Number:"<<b.account_no <<endl;
    cout<<"Balance:"<<b.balance <<endl;
}

int main(){
    BankAccount  b1(101,5000);
    BankAccount  b2(102,7500);
    BankAccount  b3(103,9000);

    cout<<"Account 1:"<<endl;
    displayAccount(b1);
    cout<<"\nAccount 2:"<<endl;
    displayAccount(b2);
    cout<<"\nAccount 3:"<<endl;
    displayAccount(b3);
}

```

```
cout<<endl;  
BankAccount::displaytotalAccounts();  
return 0;
```

OUTPUT:

```
Account 1:  
Enter Account Number:101  
Balance:5000  
  
Account 2:  
Enter Account Number:102  
Balance:7500  
  
Account 3:  
Enter Account Number:103  
Balance:9000  
  
Total accounts created:3
```